1. (a)	$f(x) = 2\sec x + 6x - 3$ $f(0.1) = -0.39 \qquad f(0.2) = 0.24$ States change of sign, continuous and hence root	M1 A1 **(2)**
(b)	Sets $f(x) = 0$, uses $\sec x = \frac{1}{\cos x}$ and makes x of $6x$ the subject $\Rightarrow 6x = 3 - \frac{2}{\cos x} \Rightarrow x = \frac{1}{2} - \frac{1}{3\cos x}$ *	B1* **(1)**
(c)	(i) $x_2 = \frac{1}{2} - \frac{1}{3\cos 0.15} = 0.16..$ $x_2 =$ awrt 0.1629 (ii) $\alpha = 0.1622$	M1 A1 A1 **(3)**
		(6 marks)

2 (a)	$10\ \text{m}^2$	B1 **(1)**
(b)	$\log_{10} 25 = 1 + 0.03T \Rightarrow T = ...$ $T = 13.26$	M1 A1cso **(2)** **(3 marks)**

3.	$\dfrac{dy}{dx} = \dfrac{4(x+3)^2 - 2(4x+1)(x+3)}{(x+3)^4}$	M1 A1
	Solves their $4(x+3)^2 - 2(4x+1)(x+3) = 0$ $\Rightarrow (x+3)(10-4x) = 0 \Rightarrow x = \ldots$	M1
	Critical value of $\dfrac{5}{2}$; Critical value of -3	A1; B1
	C increasing when $\dfrac{dy}{dx} > 0 \Rightarrow -3 < x < \dfrac{5}{2}$	A1
		(6 marks)

4 (a)

$$\frac{4x^3+2x^2+3x+8}{x^2+4} \equiv Ax+B+\frac{Cx+D}{x^2+4}$$

(i) Any correct value of $A = 4$, $B = 2$ or $C = -13$ — B1

Full method to find values for A, B and C

E.g. $4x^3+2x^2+3x+8 \equiv (Ax+B)(x^2+4)+Cx+D$

Or $x^2+0x+4 \overline{)\, 4x^3+2x^2+3x+8}$ — M1

$$\begin{array}{r} 4x+2 \\ \hline -13x \end{array}$$

$A = 4$, $B = 2$ and $C = -13$ — A1

(ii) E.g. Using $4x^3+2x^2+3x+8 \equiv (Ax+B)(x^2+4)+Cx+D$

For example $x^2 \Rightarrow B = 2$ $\quad x = 0 \Rightarrow 4\times 2 + D = 8$ $\quad D = 0$ * — B1*

(4)

(b)

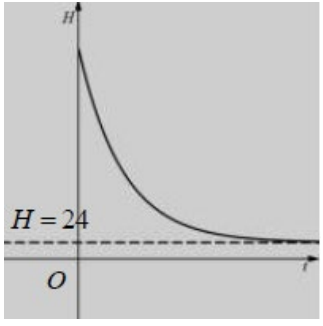
$$\int \left(Ax+B+\frac{Cx}{x^2+4} \right) \mathrm{d}x = \frac{Ax^2}{2}+Bx, +\frac{1}{2}C\ln\left(k\left(x^2+4\right)\right) \qquad k \text{ any constant}$$ — B1 ft, M1 A1ft

$$\int_1^4 \frac{4x^3+2x^2+3x-8}{x^2+4}\,\mathrm{d}x = \left[2x^2+2x-\frac{13}{2}\ln\left(k\left(x^2+4\right)\right) \right]_1^4$$

$$= \left(32+8-\frac{13}{2}\ln 20\right)-\left(2+2-\frac{13}{2}\ln 5\right) = 36+\frac{13}{2}\ln\frac{1}{4} = 36-13\ln 2$$ — dM1 A1

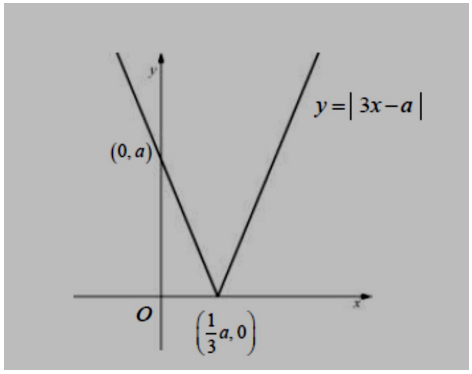
(5)

(9 marks)

5 (a)	$304(°\text{C})$	B1 **(1)**
(b)	 Shape or Asymptote Fully correct Shape and Asymptote	M1 A1 **(2)**
(c)	$144 = 280\,e^{-0.05t} + 24 \Rightarrow 280\,e^{-0.05t} = 120$ Correct use of lns $\Rightarrow -0.05t = \ln\dfrac{120}{280} \Rightarrow t = 16.95$	M1 dM1 A1 **(3)**
(d)	$\left(\dfrac{dH}{dt} =\right) -0.05 \times 280\,e^{-0.05t}$ $\dfrac{dH}{dt} = -0.05 \times (H - 24) \Rightarrow \dfrac{dH}{dt} = 1.2 - 0.05H$	M1 M1, A1cso **(3)** **(9 marks)**

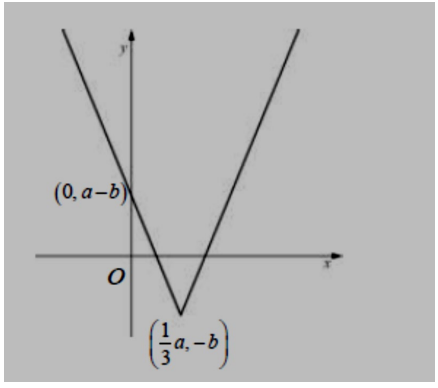
6 (a)	$y=\frac{4x+3}{x-2} \Rightarrow yx-2y=4x+3$ $\Rightarrow yx-4x=2y+3$	$y=4+\frac{11}{x-2} \Rightarrow y-4=\frac{11}{x-2}$ $x-2=\frac{11}{y-4}$ M1
	$\Rightarrow x=\frac{2y+3}{y-4} \Rightarrow \mathrm{f}^{-1}(x)=\frac{2x+3}{x-4}$	$\Rightarrow x=2+\frac{11}{y-4} \Rightarrow \mathrm{f}^{-1}(x)=2+\frac{11}{x-4}$ A1
	$x \neq 4$	B1 **(3)**
(b)	$\mathrm{ff}(x)=\frac{4\times\frac{4x+3}{x-2}+3}{\frac{4x+3}{x-2}-2}$ or $\mathrm{ff}(x)=4+\frac{11}{4+\frac{11}{x-2}-2}$	M1
	$=\frac{4\times(4x+3)+3(x-2)}{4x+3-2(x-2)}=\frac{19x+6}{2x+7}$	dM1 A1 **(3)**
(c)	Either $x=1$ or $y=38$	M1
	(1, 38)	A1 **(2)** **(8 marks)**

7 (a)(i)



Shape	B1
Vertex $\left(\frac{1}{3}a, 0\right)$	B1
y intercept $(0, a)$	B1

(a)(ii)



Shape	B1ft
Vertex $\left(\frac{1}{3}a, -b\right)$	B1ft
y intercept $(0, a-b)$	B1ft

(6)

(b)

Attempts to solve $-3x + a - b = 5x \Rightarrow x = ...$	M1
$\Rightarrow x = \frac{a-b}{8}$ ONLY	A1

(2)

(8 marks)

8 (i)	States or uses $\operatorname{cosec}\theta = \dfrac{1}{\sin\theta}$	B1
	Uses both $\operatorname{cosec}\theta = \dfrac{1}{\sin\theta}$ and $\sin 2\theta = 2\sin\theta\cos\theta$	M1
	$3\operatorname{cosec}\theta = 8\cos\theta \Rightarrow \sin 2\theta = \dfrac{3}{4}$	A1
	$\Rightarrow \theta = \dfrac{1}{2}\arcsin\left(\dfrac{3}{4}\right) =$ awrt 0.424, awrt 1.15	M1 A1
		(5)
(ii)	$\dfrac{\tan 2x - \tan 70°}{1+\tan 2x\tan 70°} = -\dfrac{3}{8} \Rightarrow \tan(2x-70°) = -\dfrac{3}{8}$	M1 A1
	Correct order of operations $x = \dfrac{\arctan\left(-\dfrac{3}{8}\right)+70°}{2}$	dM1
	awrt $24.7°$, awrt $114.7°$	A1
		(4)
		(9 marks)

9. (a)	$\frac{1}{4}$	B1 **(1)**
(b)	$f'(x) = \frac{3}{\sqrt{x}}\ln(4x) + 6\sqrt{x} \times \frac{1}{x}$	M1 A1
	Sets $\frac{3}{\sqrt{x}}\ln(4x) + 6\sqrt{x} \times \frac{1}{x} = 0 \Rightarrow \ln(4x) = -2$	dM1
	$Q\left(\frac{1}{4e^2}, -\frac{6}{e}\right)$	A1 A1 **(5)**
(c)	Attempts $-2 \times y$ co-ordinate of Q	M1
	Range $g(x)$„ $\frac{12}{e}$	A1ft **(2)**
		(8 marks)

10.(a)	$x = 3\cos 2y \Rightarrow \left(\frac{\mathrm{d}x}{\mathrm{d}y} = \right) -6\sin 2y$	M1 A1 **(2)**
(b)	E.g. $\frac{\mathrm{d}x}{\mathrm{d}y} = -6\sqrt{1-kx^2}$ or $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\text{their}\,\frac{\mathrm{d}x}{\mathrm{d}y}}$	M1
	$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{6\sqrt{1-\cos^2 2y}} = -\frac{1}{6\sqrt{1-\left(\frac{x}{3}\right)^2}} = -\frac{1}{2\sqrt{9-x^2}}$	dM1 A1 **(3)**
(c)	Sets $\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{2\sqrt{9-x^2}} = -\frac{1}{4}$	
	$x^2 = 5 \Rightarrow a = \sqrt{5}$	M1 A1
	$\sqrt{5} = 3\cos 2y \Rightarrow b = \frac{1}{2}\arccos\left(\frac{\sqrt{5}}{3}\right)$	dM1 A1
Alt (c)	Sets $\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{6\sin 2y} = -\frac{1}{4}$	
	$\Rightarrow \sin 2y = \frac{2}{3} \Rightarrow y = \frac{1}{2}\arcsin\left(\frac{2}{3}\right)$	M1A1
	$x = 3\cos 2y = 3\sqrt{1-\sin^2\left(\arcsin\frac{2}{3}\right)} = \sqrt{5}$	dM1 A1 **(4)** **(9 marks)**